

E-Waste & Environmental Management

Activity Worksheet

The Hidden Carbon Footprint of Our Gadgets

Name: Viroj Jaywant Bhadange

Roll No.: 24101A0013

Division: INF+IA.

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Value
9

Part A: Carbon Footprint Calculation

Use the carbon footprint calculator provided by your faculty.

Parameter	Your Response
Annual Carbon Footprint	20 tCO ₂ e / year
National Average	9 tCO ₂ e / year
Global Average	23 tCO ₂ e / year
Number of Earths Required (if provided)	1.2

did you learn its meaning? - Hep

Which category contributed the most?

☐ Transportation
 ☐ Home Energy
 ☒ Food
 ☐ Shopping & Consumption

☐ Electronics / Digital Lifestyle
 ☐ Other: _____

Which result surprised you the most?

I was surprised that my food choices contributed the most to my carbon footprint, showing how daily eating habits can & Annual carbon & Global Average are high for me

Part B: My Electronic Lifestyle

1. How many smartphones have you owned in the last 10 years?

☐ 1
 ☒ 2-3
 ☒ 4-5
 ☐ More than 5

2. How often do you replace your smartphone?

☐ Every year
 ☐ Every 2 years
 ☒ Every 3-4 years
 ☐ More than 5 years

3. What do you usually do with your old phone?

☒ Keep
 ☐ Sell
 ☐ Exchange
 ☐ Donate

☐ Recycle
 ☐ Throw away

4. Have you ever repaired an electronic device instead of replacing it?

☒ Yes ☐ No

If yes: I have repaired my smartphone & watch.

5. Have you ever recycled e-waste through an authorized recycler?

☐ Yes ☒ No

Item: _____

Unused Electronic Items at Home	Number
Mobile phones	<u>2</u>
Chargers	<u>1</u>
Earphones	<u>2</u>
Batteries	<u>0</u>
Cables	<u>0</u>
Other	<u>—</u>

Part C: Think Like an Environmental Engineer

Habit	Higher Carbon Footprint	More E-Waste
Buying a new phone every 2 years	☐	☒
Repairing a laptop	☐	☒
Keeping old phones unused in drawers	☐	☒
Recycling electronic devices	☒	☐
Using devices for 5–6 years	☐	☐

Which stage of an electronic product's life cycle contributes the MOST carbon emissions?

☐ Raw material extraction ☒ Manufacturing ☐ Transportation

☐ Usage ☐ Disposal

Why is extending the life of an electronic device environmentally beneficial?

It reduces

It reduces e-waste, saves raw materials & lowers carbon emissions.

Part D: Reflection

Three habits that increase my personal carbon footprint:

- Using private vehicles frequently
- Wasting electricity by leaving lights on.
- Eating processed or packaged food regularly

Three habits that increase e-waste generation:

- Buying new electronic devices frequently
- Keeping old electronics unused at home
- Not recycling old electronic devices

Action to Reduce Carbon Emissions & E-waste	Expected Impact
Uses devices for a longer time.	Reduces e-waste & carbon emissions
Repair devices instead of replacing them.	Saves resources & reduces waste
Recycle old electronics properly	Prevents pollution & recovers valuable materials.

Discussion Questions (Optional)

1. Which stage of an electronic product's life cycle has the highest carbon footprint and why?

Manufacturing, because it requires large amounts of energy & raw materials.

2. How does extending the lifespan of smartphones reduce both carbon emissions and e-waste?

It reduces the need for new phones, saving resources, lowering emissions & reducing e-waste.

3. Why is recycling alone not enough for sustainable e-waste management?

Because not all materials can be recovered, so reduce & reuse are also important.

4. Explain how the 5Rs contribute to lowering the environmental impact of electronic products.

The 5R (Refuse, Reduce, Reuse, Repair, Recycle).
help conserve resources, reduce waste,
lower pollution & decrease carbon emissions.

5. As an engineering student, what innovative solution would you propose to reduce the carbon footprint associated with electronic devices?

I would develop a smart e-waste collection
& repair platform that helps people repair
or reuse devices, & recycle electronic devices.

This would reduce e-waste, save resources
& lower carbon emissions.

Ms. Nilima Main

Assistant Professor, Department of First Year Engineering
Vidyalankar Institute of Technology